

Pricing models, conditions and the PINN

Every formula taken from the code. Notation: $x = \log(F/K)$, $\tau = T - t$, $c = C/F$, $w = \sigma_{imp}^2 \tau$, v_s, v_f = slow and fast instantaneous variance. Controlled benchmark uses $r = q = 0$, so $F = S$.

1 Black-Scholes / Black-76

$$c(x, \tau, \sigma) = \Phi(d_1) - e^{-x} \Phi(d_2), \quad d_1 = \frac{x}{\sigma \sqrt{\tau}} + \frac{\sigma \sqrt{\tau}}{2}, \quad d_2 = d_1 - \sigma \sqrt{\tau}$$

Currency price $C = Fc$. Vega (the calibration weight): $v = \partial c / \partial \sigma = \phi(d_1) \sqrt{\tau}$.

Implied volatility inverts $C/K = e^x \Phi(d) - \Phi(d - \sqrt{w})$ for w by 75 bisection steps on $w \in [10^{-12}, 40]$, then $\sigma_{imp} = \sqrt{w/\tau}$.

Inversion is defined only where $\max(e^x - 1, 0) + 10^{-12} < C/K < e^x - 10^{-12}$ — outside it the IV is NaN.

Two baselines are used: BS_FLAT, one σ for the surface; BS_TERM / BS_EXPIRY, one σ_j per calibration expiry with total variance interpolated linearly in τ and held flat beyond the knots.

2 Single Heston (5 parameters)

$$dS_t = \sqrt{v_t} S_t dW_t^S, \quad dv_t = \kappa(\theta - v_t) dt + \sigma \sqrt{v_t} dW_t^V, \quad d\langle W^S, W^V \rangle_t = \rho dt$$

3 Double Heston (10 parameters, separable four-shock)

$$dS_t = \sqrt{v_{s,t} + v_{f,t}} S_t dW_t^S$$

$$dv_{i,t} = \kappa_i(\theta_i - v_{i,t}) dt + \sigma_i \sqrt{v_{i,t}} dW_t^i, \quad d\langle W^S, W^i \rangle_t = \rho_i dt, \quad i \in \{s, f\}$$

Each variance factor carries its own correlation with the price; the two factors are independent of each other.

Per-factor characteristic exponent (Little-Heston-Trap stable form):

$$b = \kappa - \rho \sigma i u, \quad d = \sqrt{b^2 + \sigma^2(u^2 + i u)} \quad \text{with } \Re d \geq 0, \quad g = \frac{b - d}{b + d}$$

$$\psi(u, \tau) = \frac{\kappa \theta}{\sigma^2} \left[(b - d) \tau - 2 \log \frac{1 - g e^{-d \tau}}{1 - g} \right] + \frac{b - d}{\sigma^2} \cdot \frac{1 - e^{-d \tau}}{1 - g e^{-d \tau}} v_0$$

The two factors multiply — this is the whole point of the separable construction:

$$\varphi(u, \tau) = \exp[\psi_s(u, \tau) + \psi_f(u, \tau)]$$

Pricing integral (forward-normalised, undiscounted), evaluated by Gauss-Laguerre at 128 and again at 96 nodes:

$$c(x, \tau) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{i u x} \varphi(u - i, \tau)}{i u} \right] du - e^{-x} \left(\frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{i u x} \varphi(u, \tau)}{i u} \right] du \right)$$

4 Parameter admissibility

condition	formula	where enforced
positivity	$\kappa_i, \theta_i, \sigma_i, v_{0,i} > 0$	always
correlation bounds	$-1 < \rho_i < 1$	always
slow-first ordering	$\kappa_s < \kappa_f$	always (removes the label swap)
Feller, strict variant	$2\kappa_i\theta_i - \sigma_i^2 > 0$	feller_strict only
joint correlation disk	$\rho_s^2 + \rho_f^2 < 1$	reference contract, NOT the market tests

Market tests use feller_free: the characteristic function stays valid without Feller, and validation dates chose it for both Heston models (median held-out IV RMSE 3.81 vs 6.88 for Double Heston).

Under the strict variant the vol-of-vol is parameterised to satisfy Feller by construction:

$$\sigma_i = \eta_i \sqrt{2\kappa_i\theta_i} \quad \text{with } \eta_i \in [0.01, 0.999].$$

5 Numerical acceptance (else recompute by adaptive quadrature and log the fallback)

$$|c_{128} - c_{96}| \leq 10^{-7}, \quad c \geq \max(1 - e^{-x}, 0) - 10^{-9}, \quad c \leq 1 + 10^{-9}$$

6 No-arbitrage conditions, checked on every plotted curve

$$\frac{\partial C}{\partial S} \in [0, 1], \quad \frac{\partial^2 C}{\partial S^2} \geq 0, \quad \frac{\partial^2 C}{\partial K^2} \geq 0, \quad \frac{\partial w}{\partial \tau} \geq 0, \quad \max(S - K, 0) \leq C \leq S$$

and $C \rightarrow \max(S - K, 0)$ as $\tau \rightarrow 0$. Result across all 30 controlled curves: zero violations.

7 Statistical condition for a claimed win (frozen endpoint)

Double Heston beats a competitor only if BOTH hold on held-out quotes:

$$p_{\text{Wilcoxon, one-sided}} < 0.05 \quad \text{and} \quad \text{lower bound of 95\% cluster bootstrap of } \bar{\Delta} > 0$$

with $\Delta = \text{RMSE}_{\text{competitor}} - \text{RMSE}_{DH}$ averaged inside each cluster (shock episode, or calendar week), 5,000 replicates, seed 20260920.

8 The PINN output construction

The network never predicts a price. It predicts a bounded correction to an analytic baseline volatility.

$$\tilde{v}(\tau) = \sum_{i \in \{s, f\}} \left[\theta_i + (v_i - \theta_i) \frac{1 - e^{-\kappa_i \tau}}{\kappa_i \tau} \right]$$

$$\text{corr} = 1.8 \tanh(\text{head}), \quad \sigma_{\text{imp}} = \sqrt{\tilde{v}} e^{\text{corr}}, \quad w = \sigma_{\text{imp}}^2 \tau, \quad C/K = e^x \Phi(d) - \Phi(d - \sqrt{w})$$

Two properties follow by construction, not by training: the payoff at $\tau = 0$ is exact, and the price cannot leave its no-arbitrage bounds.

9 The 24 engineered features (fixed formulas, nothing learned)

With $z = x/\sqrt{\tau\bar{v} + x^2/64}$ and $\eta_i = \sigma_i/\sqrt{2\kappa_i\theta_i}$:

4 general: $x/0.36$, $z/8$, $\tanh(z/1.5)$, scaled $\log \tau$

9 per factor (18): $\log v_i$, $\log \kappa_i$, $\log \theta_i$, ρ_i , $\tanh(\kappa_i\tau)$, $v_i = \tanh(\sigma_i\sqrt{\tau}/\sqrt{\bar{v}})$, $\rho_i v_i$, $\tanh(\frac{1}{2}\log(v_i/\theta_i))$, Feller ratio $2\log(1 + \eta_i)/\log 4 - 1$

2 cross-factor: $2v_s/(v_s + v_f) - 1$, $\tanh(z/1.5) \cdot (\rho_s v_s + \rho_f v_f)$

10 Loss terms

$$L_{price} = \langle (\hat{c}_\theta - c^{teacher})^2 \rangle, \quad L_{IV} = \langle (\hat{\sigma}_\theta - \sigma^{teacher})^2 \rangle_{\text{valid IV}}$$

$$L_{PDE} = \langle \mathcal{R}^2 \rangle, \quad L_{conv} = \langle \max(-C, 0)^2 \rangle$$

$\mathcal{R}$ is the two-factor Heston PDE residual in log-total-variance form $\ell = \log w$, normalised by scale = $(v_s + v_f) + \bar{v}$, with $a = \frac{1}{2}x^2 - \frac{1}{8}w^2 - \frac{1}{2}w$:

$$\mathcal{R} = \frac{1}{\text{scale}} \left[w\ell_\tau - \frac{v_s + v_f}{2} D_x - \sum_i \rho_i \sigma_i v_i D_{xv_i} - \frac{1}{2} \sum_i \sigma_i^2 v_i D_{v_i v_i} - \sum_i \kappa_i (\theta_i - v_i) w \ell_{v_i} \right]$$

$$D_x = 2 + 2(\frac{w}{2} - x)\ell_x + a\ell_x^2 + w(\ell_{xx} + \ell_x^2) - w\ell_x$$

$$D_{xv_i} = (\frac{w}{2} - x)\ell_{v_i} + a\ell_x\ell_{v_i} + w(\ell_{xv_i} + \ell_x\ell_{v_i}), \quad D_{v_i v_i} = a\ell_{v_i}^2 + w(\ell_{v_i v_i} + \ell_{v_i}^2)$$

$$C = (1 - \frac{1}{2}x\ell_x)^2 - \frac{1}{4}w\ell_x^2 - \frac{w^2\ell_x^2}{16} + \frac{1}{2}w(\ell_{xx} + \ell_x^2) \quad (\text{convexity diagnostic})$$

All derivatives by automatic differentiation with `create_graph=True`.

11 Total loss — each term divided by its own acceptance tolerance squared

$$L = \frac{L_{price}}{(2 \times 10^{-5})^2} + w_{IV} \frac{L_{IV}}{(0.002)^2} + w_{PDE} \cdot 0.1 \cdot \frac{L_{PDE}}{(0.01)^2} + w_{conv} \cdot 0.1 \cdot L_{conv}$$

Locked v4: $w_{IV} = w_{PDE} = w_{conv} = 1$, fixed. Improved v5: updated every 100 steps by

$$w_i \leftarrow 0.9 w_i + 0.1 \cdot \min \left(\frac{\|\nabla_\theta L_{price}\|}{\|\nabla_\theta L_i\|}, 10^3 \right)$$

Residual-adaptive collocation (v5): every 2,000 steps half of the 18,000 active points are redrawn from a 120,000-point pool with $p_j \propto |\mathcal{R}_j|/|\mathcal{R}| + 1$; the other half stays uniform. Budget unchanged.

Acceptance gates (frozen before training, all four must pass): price RMSE $\leq 2 \times 10^{-5}$, $P_{95} \leq 5 \times 10^{-5}$, max $\leq 2 \times 10^{-4}$, IV RMSE ≤ 0.002 (0.2 vol points).

12 The ten Double Heston parameters (published set, slow factor first)

#	symbol	value	meaning
1	κ_s	0.9491	slow mean reversion — half-life 267 days
2	θ_s	0.0257	slow long-run variance (16.0% vol)
3	σ_s	0.0517	slow vol-of-vol
4	ρ_s	+0.7009	slow price/variance correlation
5	$v_{s,0}$	0.0003	slow initial variance (1.7% vol)
6	κ_f	10.7526	fast mean reversion — half-life 24 days
7	θ_f	0.0330	fast long-run variance (18.2% vol)
8	σ_f	0.3613	fast vol-of-vol
9	ρ_f	-0.8916	fast price/variance correlation
10	$v_{f,0}$	0.0252	fast initial variance (15.9% vol)

Source: Christoffersen, Heston & Jacobs (2009).

Structure: $\kappa_f/\kappa_s = 11.33$ — two clearly separated timescales; and $\rho_s > 0 > \rho_f$ — opposite-signed correlations, the mechanism behind short-dated smirk flexibility and the reason the separable four-shock convention is required.

$$v_{s,0} + v_{f,0} = 0.0255 \Rightarrow \sigma_{BS, fixed} = 15.97\%, \quad \theta_s + \theta_f = 0.0587 \Rightarrow 24.2\% \text{ long-run}$$

Feller: slow $2\kappa_s\theta_s - \sigma_s^2 = 0.0461 > 0$, fast $0.5791 > 0$ — both pass.

Joint disk: $\rho_s^2 + \rho_f^2 = 1.286 > 1$ — the published set fails the stricter repository convention, which is why the market tests use the individual- ρ contract.

13 Published Single Heston (5 parameters)

$$(\kappa, \theta, \sigma, \rho, v_0) = (8.9814, 0.0409, 0.2970, -0.9621, 0.0244), \quad \text{Feller} = 0.6465 > 0$$

14 The level scale s — the only number ever fitted to a fixed-parameter model

$$\theta_i \mapsto s\theta_i, \quad \sigma_i \mapsto \sqrt{s}\sigma_i, \quad v_{i,0} \mapsto s v_{i,0}, \quad \sigma_{BS} \mapsto \sqrt{s}\sigma_{BS}, \quad s \in [0.7, 3.2]$$

Rescales the volatility LEVEL only; every timescale, correlation and shape is untouched. The range is exactly the range the PINN was trained on.

15 Controlled scenario FIXED_TOTAL_TWIST

state	$v_{f,0}$	$v_{s,0}$	total	instantaneous vol
FAST-HEAVY	0.035	0.005	0.04	20%
SLOW-HEAVY	0.005	0.035	0.04	20%

Identical instantaneous volatility, materially different prices at every maturity, because variance in the fast factor decays 11.3x faster. No single-factor model can produce both curves from the same variance.

16 PINN training domain — outside this box the network is untrained

variable	range	sampling
$x = \log(F/K)$	$[-0.36, +0.36]$ i.e. $S/K \in [0.698, 1.433]$	uniform
τ	7 days to 2 years (0.0192 - 2.0 y)	uniform in $\log \tau$
v_s slow variance	$[1.5 \times 10^{-4}, 0.18]$ i.e. 1.2% - 42% vol	uniform in $\log v_s$
v_f fast variance	$[1.0 \times 10^{-3}, 0.22]$ i.e. 3.2% - 47% vol	uniform in $\log v_f$
level scale s	$[0.7, 3.2]$	uniform
sampler	5-dimensional Latin hypercube, one seed per split	

Datasets: teacher train 100,000 (seed 93101); collocation 18,000 (93103); v_5 development 8,192 (93201); v_5 UNTOUCHED final 16,384 (93202); PDE final 4,096 (93203); short-ATM diagnostic 6,144 (93204); RAD pool 120,000 (93205).

17 Calibration bounds (market tests) and objective

parameter	bound	parameter	bound
κ (single)	[0.05, 50]	σ_i	[0.01, 10]
κ_s	[0.05, 20]	ρ_i	$[-0.99, 0.99]$
$\kappa_f - \kappa_s$ (gap)	[0.05, 100]	η_i (strict Feller)	[0.01, 0.999]
θ_i	[0.001, 4]	σ_{BS}	[0.05, 5]
$V_{i,0}$	[0.001, 4]		

$$\min \left\langle \left(\frac{C_{model} - C_{market}}{\max(v, 10^{-4})} \right)^2 \right\rangle$$

Dividing by vega makes this approximately an implied-volatility residual. Optimiser: differential evolution (popsize x6, 25 iterations, seed 20260919) then 12 bounded least-squares starts, all starts retained.

18 Market data filters

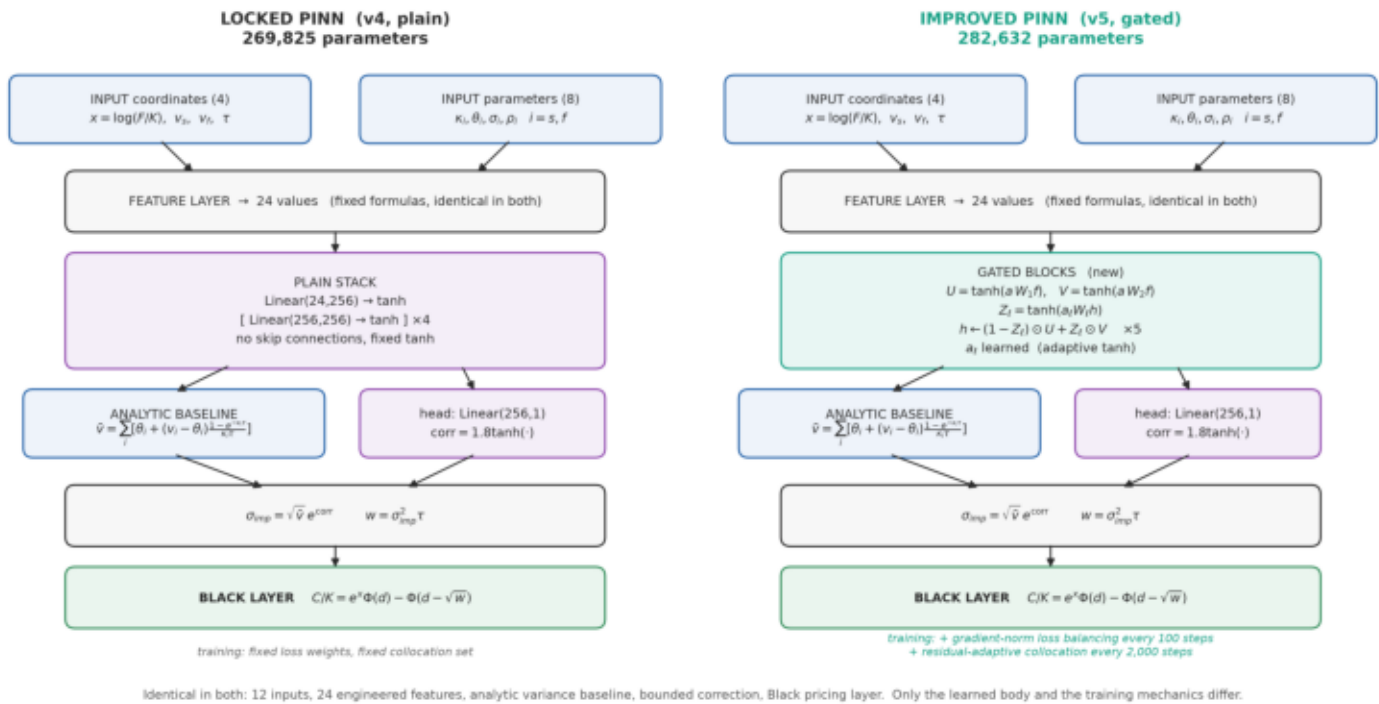
	crypto (Deribit)	equity / index (CBOE)
maturity	3 - 400 days	7 - 730 days
moneyness	$ x \leq 1.0$	$ x \leq 0.36$ (the PINN domain)
implied vol	0.10 - 3.00	0.02 - 3.00
option type	out-of-the-money only	out-of-the-money only
minimum price	0.0005 coin / 2 ticks	bid > 0, ask > bid, ask ≤ 3 bid
quotes per expiry	≥ 3	≥ 6
expiries per date	≥ 5 (BTC/ETH), ≥ 4 thin	≥ 6
window	06:00-08:00 UTC or full day	single snapshot
forward	solved per trade from (price, IV)	parity regression, $R^2 \geq 0.999$

19 Range of quantities plotted

quantity	range
S (controlled)	0.70K to 1.30K with $K = 100$; ATM zoom 0.85K - 1.15K
τ	7 - 730 days, 300-500 grid points; panels at 30, 90, 180, 365, 730 d
calendar time t	0 to T with $\tau = T - t$; PINN curves stop at $\tau = 7$ d
$c = C/F$	$[0, 1)$ by construction
time value	0 to about 4 at 90 days ($K = 100$)
implied volatility	about 15-26% controlled, 9-50% market

20 Block architecture: locked v4 and improved v5

Double Heston PINN: prior-plus-correction architecture, before and after



Identical in both: 12 inputs, 24 engineered features, analytic variance baseline, bounded correction, Black pricing layer. Only the learned body and the training mechanics differ.

21 Measured effect, untouched 16,384-point test set opened once

metric	locked v4	improved v5	change
price RMSE vs exact DH	1.062e-5	6.610e-6	1.61x better
IV RMSE (vol points)	0.1175	0.0738	1.59x better
worst-case price error	1.523e-4	5.938e-5	2.56x better
PDE residual	0.0309	0.0558	1.8x WORSE
parameters	269,825	282,632	+4.7%
inference latency	14.2 us/quote	24.8 us/quote	1.7x slower

The improvement is a better numerical solver, not a better market model: it moved real SPX pricing error by 0.003 vol points out of 5.8, because network error is about 1% of model error.

22 Where each equation lives in the code

equation	file
Black-76 price, vega, implied vol	experiments/btc_multifactor_v1/engine.py
IV inversion and validity window	src/mentor_dh_pinn/regular_pinn_data.py
factor characteristic exponent	src/double_heston_reference.py::factor_exponent
Fourier pricing integral	experiments/btc_multifactor_v1/engine.py::Grid
admissibility, Feller, disk	engine.py::admissible, src/constraints.py
baseline, features, correction, PDE	src/mentor_dh_pinn/regular_pinn_torch.py
loss, gradient balancing, RAD	experiments/dh_pinn_v5/train.py
architecture variants	experiments/dh_pinn_v5/v5_models.py
published parameters, domains	experiments/nifty_multifactor_v4/config.json
calibration bounds and objective	experiments/btc_multifactor_v1/config.json